

# CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

## University Examination

Date: 9.05.2018

Day: Wednesday

Time: 10.00 am to 10.30 am

Total Marks: 20

|                           |  |  |  |  |  |  |  |  |  |
|---------------------------|--|--|--|--|--|--|--|--|--|
| Student's ID No.          |  |  |  |  |  |  |  |  |  |
|                           |  |  |  |  |  |  |  |  |  |
| Student's Name: _____     |  |  |  |  |  |  |  |  |  |
| Student's Sign : _____    |  |  |  |  |  |  |  |  |  |
| Supervisor's Name : _____ |  |  |  |  |  |  |  |  |  |
| Supervisor's Sign: _____  |  |  |  |  |  |  |  |  |  |

|                                                                                          |
|------------------------------------------------------------------------------------------|
| Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following |
| Examination : BPT – May 2018                                                             |
| Semester: [8]                                                                            |
| Section : I                                                                              |
| Course Code : PT408.01                                                                   |
| Course Name: ERGONOMICS                                                                  |

### Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes. Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

### SECTION-I

#### Multiple Choice Questions: (MCQs) (Encircle the correct answer)

(10 × 1 = 10)

1. Which of the following is not a Cumulative Trauma Disorder (CTD)?
  - a. Periarthritis.
  - b. Tenosynovitis.
  - c. Carpel tunnel syndrome
  - d. Tennis Elbow
2. Which of the following does not optimize lifting conditions?
  - a. Holding the load close to the body.
  - b. Picking up the load with both hands.
  - c. Twisting the trunk when lifting.
  - d. Lifting loads with handles.

3. The Ergonomics society was formulated in the year .....
  - a. 1949
  - b. 1950
  - c. 1951
  - d. 1952
4. Leisure ergonomics started from .....
  - a. 1970
  - b. 1980
  - c. 1990
  - d. 2000
5. What are the four reference postures for computer work stations according to the Human Factors and Ergonomics Society?
  - a. Reclined sitting, declined sitting, kneeling, standing on a fatigue mat
  - b. Reclined sitting, upright sitting, declined sitting, standing
  - c. Sitting on the floor, sitting on a chair, kneeling, standing
  - d. Relaxed sitting, upright sitting, kneeling, standing on a fatigue mat
6. Why do ergonomists use task analysis?
  - a. To gain an understanding of what people do in the jobs they carry out
  - b. To identify job errors and improve job satisfaction
  - c. To make jobs more efficient
  - d. To discover interface issues
7. The viewing angle of monitors is..... below horizontal.
  - a. 15-20 degree
  - b. 5-10 degree
  - c. 10-15 degree
  - d. 20- 30 degree
8. What is the best approach to address the poor worker interaction issues in that workplace?
  - a. Akamiso training
  - b. Re-training with a focus on safety
  - c. Participatory ergonomics
  - d. Organizational assessment
9. What is reflectance?
  - a. The amount of light reflected by a surface
  - b. The rate at which luminous energy is generated by a light source
  - c. The amount of light falling on an area
  - d. The proportion of light reflected by a lighted surface
10. The greater the match between worker capabilities and task demands:
  - a. The greater the chance of worker injury/illness
  - b. The lower the chance of worker injury/illness
  - c. Has no impact on chance of worker injury/illness
  - d. The greater the chance of being exposed to high risk jobs

**Identify the sentence as True or False:**

**(5 × 1 = 5)**

**(Please tick mark the answer)**

**True    False**

**11.** The 8 hour exposure of noise is 80 decibel.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**12.** Eye strain cannot be prevented by task rotation.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**13.** To relieve stress and repetative strain injuiry, brief breaks should be taken  
Every 30 minutes.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**14.** The exerted manual force for pushing and pulling is 30 kg.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**15.** The correct arm position at your computer helps relieve tension in your  
neck and shoulders.

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
|--------------------------|--------------------------|

**Match the following**

**(5 × 1 = 5)**

|            |                                                           |                 |
|------------|-----------------------------------------------------------|-----------------|
| <b>16.</b> | Maximum load for lifting should not be more than          | <b>A.</b> 25cm  |
| <b>17.</b> | Vertical displacement of the load                         | <b>B.</b> 23kg  |
| <b>18.</b> | Amount of energy required during running                  | <b>C.</b> 33kg  |
| <b>19.</b> | If trolley moves more than one min, then exerted force is | <b>D.</b> 670W  |
| <b>20.</b> | Amount of energy required during rest                     | <b>E.</b> 250 W |
|            |                                                           | <b>F.</b> 10 kg |
|            |                                                           | <b>G.</b> 35cm  |

(16 -    ),      (17-    ),      (18-    ),      (19-    ),      (20-    )

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## VIII<sup>th</sup> Semester of B.P.T. Examination

May 2018

### PT408.01 ERGONOMICS

Date: 9.05.2018

Time: 10.30 am to 1.00 pm

Maximum Marks: 50

#### Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

#### SECTION-II

#### Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. Define Ergonomics by IEA.
- b. Any four factors affecting posture.
- c. Ergonomical aspects of illumination.
- d. List out the Poor postural habits.
- e. Biomechanical principles of ergonomics.
- f. How do you avoid MSDs?
- g. What is an ideal posture?

#### Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- a. Ergonomical aspects of noise.
- b. Ergonomical aspects of hand and arm posture.
- c. Contributing factors of ergonomics.
- d. Ergonomical aspects of lifting.
- e. Work organisation

#### Q. 3. Essay (2 out of 3) (2 × 10 = 20)

- a. What are the risk factors of WRMSDs? What are the common problems leading to WRMSDs and explain how will you prevent them?
- b. Describe the ergonomical aspects of work station design for the computer users.
- c. Explain the principles, disciplines and benefits of ergonomics.